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Postradiation Optic Atrophy Is Associated With Intraocular Pressure and May Manifest With Neuroretinal Rim Thinning

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Background: To determine risk factors for postradiation optic atrophy (PROA) after plaque radiotherapy for uveal melanoma.
Methods: A single center, retrospective cohort study of patients diagnosed with uveal melanoma involving choroid and/or ciliary body treated with plaque between January 1, 2008, and December 31, 2016. Outcomes included development of PROA with pallor alone or with concomitant neuroretinal rim thinning (NRT). Cox regression analysis was performed to identify risk factors for PROA.
Results: Of 78 plaque-irradiated patients, PROA developed in 41 (53%), with concomitant NRT in 15 (19%). Risk factors for PROA of any type included presentation with worse visual acuity (odds ratio [95% confidence interval] 5.6 [2.3–14.1], $P < 0.001$), higher baseline intraocular pressure (IOP; 14 vs 16 mm Hg) (1.1 [1.0–1.2], $P = 0.03$), shorter tumor distance to optic disc (1.3 [1.2–1.5], $P < 0.001$) and foveola (1.2 [1.1–1.3], $P < 0.001$), subfoveal subretinal fluid (3.8 [2.0–7.1], $P < 0.001$), greater radiation prescription depth (1.3 [1.1–1.6], $P = 0.002$), dose to fovea (point dose) (1.01 [1.01–1.02], $P < 0.001$), and mean (1.02 [1.02–1.03], $P < 0.001$) and maximum dose to optic disc per 1 Gy increase (1.02 [1.01–1.03], $P < 0.001$). On multivariate modeling, dose to disc, baseline IOP, and subfoveal fluid remained significant. Subanalysis revealed risk factors for pallor with NRT of greater mean radiation dose to disc (1.03 [1.01–1.05], $P = 0.003$), higher maximum IOP (17 vs 20 mm Hg) (1.4 [1.2–1.7], $P < 0.001$), and subfoveal fluid (12 [2–63], $P = 0.004$).
Conclusion: PROA may result in NRT in addition to optic disc pallor. Risk factors for PROA included higher radiation dose to optic disc, higher baseline IOP, and subfoveal fluid. Higher maximum IOP contributed to concomitant NRT.

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Radiation-based therapies, including plaque, proton beam, and gamma knife, are safe and effective modalities to treat uveal melanoma (1–6). However, radiation side effects limit long-term visual acuity (7–9). Although numerous studies have focused on prevention and treatment of radiation maculopathy and retinopathy (8,10–13), fewer studies have been dedicated to studying postradiation optic atrophy (PROA). Kim et al (14) reported 93 patients treated with proton beam irradiation for parapapillary uveal melanoma and found that 68% developed PROA at a median of 1.5 years after treatment. Despite the high radiation dose to optic disc associated with the treatment of parapapillary melanoma, nearly a third of patients exhibited no measurable evidence of PROA. Shields et al (8), in a series of 1,248 patients treated with plaque radiotherapy for uveal melanoma in any location, found PROA in 17%–27% of patients after 21–27 months. Radiation dose to the optic disc is a well-established risk factor for PROA and is associated with a poor visual acuity outcome (9,14–16).

In our practice, we observed that over the course of several years, some cases of PROA displayed optic disc pallor as the predominant feature, whereas others developed concomitant neuroretinal rim thinning (NRT). In this article, we explore the clinical features and risk factors of

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PROA and explore an association with NRT and intraocular pressure (IOP).

METHODS

A retrospective chart review was performed on a subset of patients enrolled in the Prospective Ocular Tumor Study (POTS) at Mayo Clinic, Rochester, MN, between July 15, 2019, and February 25, 2020. Included patients were diagnosed with uveal melanoma involving the choroid and/or ciliary body and treated with plaque radiotherapy between January 1, 2008, and December 31, 2016. Patients were required to have a minimum of 3-year follow-up after plaque and at least 2 fundus photographs containing a clear, unobstructed view of the optic disc, one before plaque and one ≥3 years after treatment. Patients with isolated iris melanoma, <3-year follow-up, or lack of clear photographs of the optic disc were excluded. This study complied with the Health Insurance Portability and Accountability Act and adhered to the tenets of the Declaration of Helsinki. The Mayo Clinic Institutional Review Board deemed the study exempt as a retrospective chart review only, and informed consent was obtained from all patients for inclusion in the prospectively Institutional Review Board–approved POTS database.

All patients underwent a complete eye examination by an ocular oncologist (L.A.D. and T.W.O.) with color fundus drawing, fundus photography, autofluorescence, A-scan and B-scan ultrasonography, fluorescein angiography, and optical coherence tomography (OCT) as indicated. Plaque radiotherapy was performed under anesthesia using I-125 Collaborative Ocular Melanoma Study (COMS) plaques with tumor localization using transillumination and/or indirect ophthalmoscopy. Intraoperative ultrasonography was used to confirm proper plaque placement, equally overlapping all tumors' margins when possible or offset if tumor was abutting the optic disc. Planned treatment duration was approximately 94 hours. Records were reviewed for patient demographics, clinical tumor features, treatment parameters, and outcomes, with attention to IOP and optic disc features.

PROA was defined as the initial development of optic disc pallor with or without NRT. Because optic disc edema is associated with the development of PROA, in cases where disc edema was observed at a routine visit, this was defined as the initial presentation of PROA. Sectoral disc pallor associated with retinal vein occlusion was not counted as PROA. Disc color over time was compared with baseline optic disc color of the treated eye and disc color of the fellow eye at each matching time point. Patients who underwent cataract surgery in only 1 eye were excluded because this prevented adequate comparison of disc color between the treated and fellow eye. Patients treated with transpupillary thermotherapy within 6 mm of the optic disc and those who developed diffuse retinal atrophy were also excluded to avoid confounding factors that could contribute to optic atrophy. Detailed optic disc data were collected from photographic review before treatment and after treatment when available at 6, 12, 24, 36, 48, 60, and 120 months, including IOP, cup-to-disc ratio, inferior and superior optic disc rim (expressed as percentage of total disc vertical height), Drance hemorrhage, disc edema, diffuse pallor, neovascularization of the disc (NVD), and treatment with IOP-lowering therapy. All images were graded by a single observer (L.A.D.) who was masked to all clinical data. Optic disc rim was measured using calipers at the 12-o'clock and 6-o'clock positions. Any hesitations by the grader (L.A.D.) were addressed by consensus review with a glaucoma specialist (G.W.R.).

Radiation dosages to the tumor, fovea, and optic disc were calculated using EyeDose, an open source MATLAB program developed by Deufel et al. The program uses Monte Carlo methods to calculate 3-dimensional heterogeneity-corrected dose distributions for I-125 COMS plaques and has been validated against published Monte Carlo results for 8 different tumor positions.

Statistical analysis was performed using SPSS Statistics Software version 22 (IBM, Armonk, NY). Demographics, clinical features, treatment features, and outcomes were compared for eyes with no PROA, PROA with pallor alone, and PROA with pallor plus NRT. Categorical variables were compared using the Fisher exact test, and continuous variables were compared using the Kruskal–Wallis H test. A subanalysis was performed to compare the eyes affected by PROA manifesting as pallor alone vs pallor plus NRT. Categorical variables were compared using the Fisher exact test, and continuous variables were compared using the Mann–Whitney *U* test. Cox regression analysis was used to determine risk factors for PROA by univariate and multivariate analysis using the stepwise method. The Benjamini–Hochberg procedure was used to control the false discovery rate for multiple comparisons using a q value of 0.10. A P value of <0.05 was considered statistically significant.

RESULTS

There were 78 eyes of 78 patients included in the study. Of these, 37 (47%) developed no PROA and 41 (53%) developed PROA, with pallor alone in 26 (33%) or pallor and NRT in 15 (19%) (Fig. 1). Patient demographics are listed in Table 1, and a comparison (no PROA vs pallor vs pallor and NRT) revealed no differences between groups.

Clinical features are listed in Table 2. A comparison (no PROA vs pallor vs pallor and NRT) revealed eyes with PROA of any type presented with worse logMAR visual acuity ($P < 0.001$), shorter tumor distance to optic disc ($P < 0.001$) and foveola ($P < 0.001$), and more frequent subfoveal fluid ($P < 0.001$).

Treatment features are listed in Table 3. A comparison (no PROA vs pallor vs pallor and NRT) revealed eyes with PROA of any type were treated with greater prescription depth ($P = 0.04$), greater radiation dose to fovea (point dose)

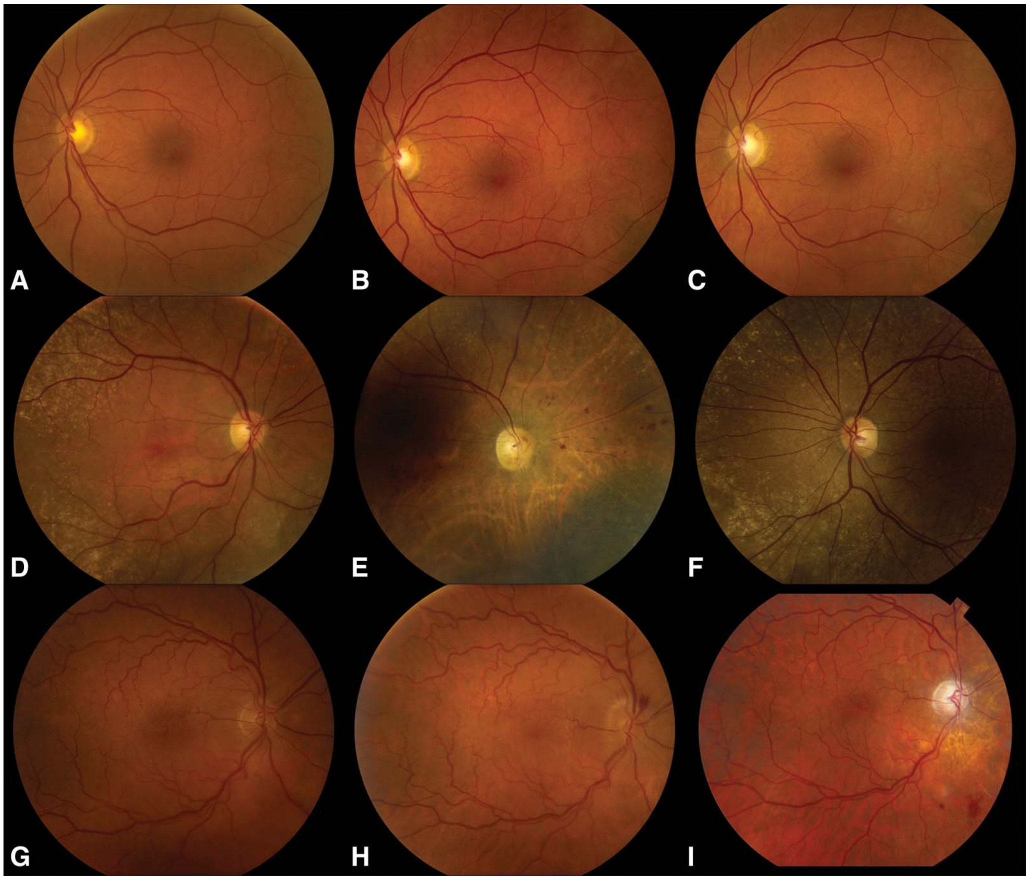

FIG. 1. Uveal melanoma treated with plaque radiotherapy with no PROA, with pallor alone, and PROA with pallor and NRT. A 73-year-old woman was treated for choroidal melanoma in the temporal periphery from the equator or ora serrata. No optic neuropathy was present (**A**) before plaque radiotherapy, (**B**) 36 months after radiotherapy, or (**C**) 60 months after radiotherapy. A 79-year-old woman was treated for choroidal melanoma in the inferonasal quadrant. No optic neuropathy was present (**D**) before plaque radiotherapy. **E**. Optic disc pallor was noted 36 months after treatment compared with (**F**) the normal fellow eye. A 54-year-old man was treated for choroidal melanoma in the nasal midperiphery from the macula to equator. No optic neuropathy was present (**G**) before plaque radiotherapy. **H**. Optic disc edema and Drance hemorrhage were noted 12 months after treatment, with (**I**) progression to optic disc pallor with NRT after 36 months. NRT, neuroretinal rim thinning; PROA, postradiation optic atrophy.

($P < 0.001$), greater mean radiation dose to optic disc ($P < 0.001$), and greater maximum radiation dose to optic disc ($P < 0.001$). No specific treatment was recommended for PROA, although 74% of all study eyes received intravitreal anti-vascular endothelial growth factor (anti-VEGF) at some point for radiation retinopathy and/or maculopathy.

Clinical outcomes are listed in Table 4. A comparison (no PROA vs pallor vs pallor and NRT) revealed eyes with both pallor and NRT had longer follow-up time ($P = 0.01$). The eyes with PROA of any type had worse final logMAR visual acuity ($P < 0.001$), greater frequency of radiation maculopathy ($P = 0.01$), nonproliferative radiation retinopathy ($P = 0.003$), branch retinal vein occlusion ($P < 0.001$), and proliferative radiation retinopathy ($P = 0.02$). There were no cases of neovascular glaucoma. There was no difference in time to PROA by PROA subtype (pallor alone vs pallor and NRT) (29 [23] vs 30 [22] months, $P = 0.59$).

Detailed optic disc features over time are listed in **Supplemental Digital Content 1** (see **Table**, http://links.lww.com/WNO/A535). No eyes had baseline glaucoma, Drance hemorrhage, disc edema, pallor, or NVD, and no eyes were being treated with IOP-lowering therapy. Eyes with any type of PROA had greater frequency of Drance hemorrhage ($P = 0.001$). Eyes with pallor alone had the greatest frequency of disc edema ($P < 0.001$), and eyes with pallor and NRT had the most frequent requirement for topical IOP-lowering therapy ($P = 0.002$). One eye in the pallor and NRT group had a single time point spike in IOP to 33 mm Hg at 12 months after plaque, which responded to temporary use of topical aqueous suppression. There were no other instances of IOP higher than 25 mm Hg. No fellow eyes developed ocular hypertension, glaucomatous optic neuropathy, or nonglaucomatous optic neuropathies during the study period. Specifically, no fellow, nonradiated eyes developed NRT, whereas 19% (15/78) did in the treated

TABLE 1. Risk factors for postradiation optic atrophy after plaque radiotherapy for uveal melanoma

Demographics	No Postradiation Optic Atrophy (n = 37) [N (%)]	Pallor Only (n = 26) [N (%)]	Neuroretinal Rim Thinning and Pallor (n = 15) [N (%)]	*P* Value	*P* Value*	Total (N = 78) [N (%)]
Age (yr)						
Mean (median, range)	56 (56, 28–87)	60 (64, 23–80)	56 (59, 28–69)	0.09	0.05	56 (60, 23–87)
Race						
Caucasian	37 (100)	25 (96)	15 (100)	0.53	0.99	77 (99)
Hispanic	0 (0)	1 (4)	0 (0)			1 (1)
Sex						
Male	9 (24)	12 (46)	7 (47)	0.14	0.99	28 (36)
Female	28 (76)	14 (53)	8 (53)			50 (64)
Involved eye						
Right	18 (49)	15 (58)	7 (47)	0.76	0.53	40 (51)
Left	19 (51)	11 (42)	8 (53)			38 (49)
Medical history						
Diabetes	1 (3)	3 (12)	0 (0)	0.25	0.29	4 (5)
Hypertension	11 (30)	10 (38)	3 (20)	0.50	0.30	24 (31)
Smoking history	10 (27)	9 (35)	5 (33)	0.81	0.99	24 (31)
Ocular history						
Ocular melanocytosis	0 (0)	0 (0)	0 (0)	NA	NA	0 (0)
Pre-existing glaucoma	0 (0)	0 (0)	0 (0)	NA	NA	0 (0)

Demographics.
Bold values indicate a significant *P* value. The initial *P* value compares all 3 groups.
*Second *P* value directly compares the latter 2 groups with optic neuropathy characterized by pallor alone or pallor with additional neuroretinal rim thinning.
NA, not applicable.

group ($P < 0.001$). On subanalysis comparing PROA eyes with pallor alone vs pallor and NRT, eyes with both pallor and NRT had greater maximum IOP after plaque (P = 0.04), but there was no difference in the overall frequency of Drance hemorrhage or optic disc edema.

Factors associated with PROA on univariate and multivariate logistic regression analysis are listed in Table 5. A primary analysis was run for patients with any type of PROA (pallor with or without NRT), and a model was constructed for multivariate analysis using objective factors that were significant on univariate analysis, including IOP at presentation, subfoveal subretinal fluid, radiation prescription depth, radiation dose to fovea, and mean radiation dose to optic disc. Significant predictors of all type PROA on multivariate analysis were greater mean radiation dose to optic disc (P = 0.001), higher baseline IOP (P = 0.01), and subfoveal subretinal fluid (P = 0.05). A subanalysis was performed to examine risk factors specific to the development of concomitant NRT and optic disc pallor. A model was constructed for multivariate analysis using objective factors that were significant on univariate analysis, including higher maximum IOP after plaque, subfoveal subretinal fluid at presentation, radiation prescription depth, radiation dose to fovea, and mean radiation dose to optic disc. Significant predictors of PROA with NRT on multivariate analysis were greater mean radiation dose to optic disc (P = 0.003), higher maximum IOP ($P < 0.001$), and subfoveal subretinal fluid (P = 0.004). All significant variables remained significant after controlling for the false discovery rate.

DISCUSSION

PROA is a well-recognized side effect after radiotherapy for uveal melanoma (7–9). In 2006, Shields et al (17) reported a series of 9 patients treated with intravitreal triamcinolone for PROA and found stable or improved visual acuity in 7 patients (78%), with resolution of disc edema in all patients, after the mean follow-up of 11 months. In 2018, Roelefs et al (18) reported a series of 11 patients, of which 9 were treated with intravitreal bevacizumab and triamcinolone. Of 4 patients presenting with decreased visual acuity, all (100%) had improvement in vision, and of 5 patients presenting without change in vision, 4 (80%) maintained stable acuity (18). In 2019, Eckstein et al (19) reported 93 patients, of which 45 were treated with intravitreal bevacizumab, triamcinolone, and/or dexamethasone, and found no difference in frequency or timing of optic atrophy or visual acuity outcome compared with observation alone.

These treatment modalities might be administered too late to prevent lasting nerve damage, or the trialed medications might not accurately target the root cause of optic nerve injury because little is known about the molecular and structural events that lead to changes seen at the optic nerve head. In this study, we investigated the clinical course of and risk factors for PROA to identify

TABLE 2. Risk factors for postradiation optic atrophy after plaque radiotherapy for uveal melanoma

Clinical Features	No Postradiation Optic Atrophy (n = 37) [N (%)]	Pallor Only (n = 26) [N (%)]	Neuroretinal Rim Thinning and Pallor (n = 15) [N (%)]	*P* Value	*P* Value*	Total (N = 78) [N (%)]
Snellen visual acuity						
20/20–20/40	37 (100)	20 (77)	8 (53)	<**0.001**	0.17	65 (83)
20/50–20/150	0 (0)	5 (19)	6 (40)	<**0.001**	0.27	11 (14)
20/200 or worse	0 (0)	1 (4)	1 (7)	0.27	0.99	2 (3)
Visual acuity (Snellen equivalent), mean (median, range)	20/25 (20/20, 20/20–20/40)	20/40 (20/25, 20/20–20/200)	20/50 (20/40, 20/20–20/200)	<**0.001**	0.18	20/30 (20/25, 20/20–20/200)
Visual acuity (logMAR), mean (median, range)	0.06 (0.00, 0.00–0.30)	0.24 (0.10, 0.00–1.00)	0.36 (0.30, 0.00–1.00)			0.18 (0.10, 0.00–1.00)
Tumor features						
Distance to optic disc (mm), mean (median, range)	8.2 (7.0, 2.0–17.0)	4.4 (3.0, 1.0–15.0)	4.0 (4.0, 0.0–15.0)	<**0.001**	0.70	6.1 (5.0, 0.0–17.0)
Distance to foveola (mm), mean (median, range)	7.1 (8.0, 0.0–15.0)	3.5 (2.5, 0.0–13.0)	3.2 (2.0, 0.0–15.0)	<**0.001**	0.94	5.2 (4.0, 0.0–15.0)
Tumor basal diameter (mm), mean (median, range)	11.8 (12.0, 7.0–20.0)	11.9 (13.0, 5.0–18.0)	12.5 (13.0, 6.0–18.0)	0.78	0.68	11.9 (12.5, 5.0–20.0)
Tumor thickness (mm), mean (median, range)	3.5 (3.2, 0.9–7.1)	4.1 (3.9, 1.4–8.3)	4.7 (4.2, 1.0–11.7)	0.43	0.98	3.9 (3.5, 0.9–11.7)
Bruch membrane rupture	0 (0)	2 (8)	2 (13)	0.06	0.61	4 (5)
Sentinel vessels	1 (3)	1 (4)	1 (7)	0.78	0.99	3 (4)
Extraocular extension	0 (0)	0 (0)	0 (0)	NA	NA	0 (0)
Tumor epicenter						
Macula	5 (14)	14 (54)	7 (47)	**0.001**	0.75	26 (33)
Macula to equator	29 (78)	11 (42)	7 (47)	**0.01**	0.99	47 (60)
Equator to ora serrata	3 (8)	0 (0)	0 (0)	0.29	NA	3 (4)
Ciliary body	0 (0)	1 (4)	1 (7)	0.27	0.99	2 (3)
Tumor configuration by ultrasound						
Dome	35 (95)	23 (88)	12 (80)	0.27	0.65	70 (90)
Mushroom	0 (0)	2 (8)	1 (7)	0.19	0.99	3 (4)
Bilobed or multilobed	2 (5)	1 (4)	2 (13)	0.58	0.54	5 (6)

(Continued)

Clinical Features	No Postradiation Optic Atrophy (n = 37) [N (%)]	Pallor Only (n = 26) [N (%)]	Neuroretinal Rim Thinning and Pallor (n = 15) [N (%)]	*P* Value	*P* Value*	Total (N = 78) [N (%)]
Subretinal fluid quadrants						
Absent	9 (24)	2 (8)	1 (7)	0.17	0.99	12 (15)
1 quadrant	25 (68)	17 (65)	8 (53)	0.61	0.52	50 (64)
2 quadrants	3 (8)	6 (23)	5 (33)	0.07	0.72	14 (18)
3 quadrants	0 (0)	1 (4)	1 (7)	0.27	0.99	2 (3)
4 quadrants	0 (0)	0 (0)	0 (0)	NA	NA	0 (0)
Subfoveal subretinal fluid by OCT						
Absent	36 (97)	16 (62)	6 (40)	**<0.001**	0.21	58 (74)
Present	1 (3)	10 (38)	9 (60)			20 (26)
Orange pigment by autofluorescence						
Absent	17 (46)	9 (35)	5 (33)	0.62	0.99	31 (40)
Present	20 (54)	17 (65)	10 (67)			47 (60)

Clinical features.
Bold values indicate a significant *P* value. The initial *P* value compares all 3 groups.
*Second *P* value directly compares the latter 2 groups with optic neuropathy characterized by pallor alone or pallor with additional neuroretinal rim thinning.
NA, not applicable; OCT, optical coherence tomography.

TABLE 3. Risk factors for postradiation optic atrophy after plaque radiotherapy for uveal melanoma

Treatment Features	No Postradiation Optic Atrophy (n = 37) [N (%)]	Pallor Only (n = 26) [N (%)]	Neuroretinal Rim Thinning and Pallor (n = 15) [N (%)]	*P* Value	*P* Value*	Total (N = 78) [N (%)]
Plaque shape						
Round	37 (100)	26 (100)	15 (100)	NA	NA	78 (100)
Plaque size						
Size (mm), mean (median, range)	17 (16, 14–22)	18 (18, 14–22)	18 (18, 14–22)	0.16	0.93	18 (18, 14–22)
Radiation features						
Prescription dose (Gy), mean (median, range)	81 (85, 60–86)	78 (85, 60–85)	81 (85, 65–85)	0.60	0.57	80 (85, 60–86)
Prescription depth (mm), mean (median, range)	5 (5, 3–9)	6 (5, 4–9)	6 (5, 2–12)	**0.04**	0.86	5 (5, 2–12)
Prescription time (h), mean (median, range)	94 (94, 94–94)	94 (94, 94–94)	94 (94, 94–94)	NA	NA	94 (94, 94–94)
Total radiation dose to tumor apex (Gy), mean (median, range) [5th–95th percentile]	97 (91, 58–148) [74–130]	92 (89, 74–118) [74–116]	95 (89, 76–136) [77–131]	0.67	0.87	95 (90, 58–148) [74–129]
Radiation dose rate to tumor apex (Gy/h), mean (median, range) [5th–95th percentile]	1.04 (0.96, 0.62–1.58) [0.79–1.38]	0.98 (0.94, 0.79–1.26) [0.79–1.23]	1.01 (0.95, 0.81–1.45) [0.82–1.39]	0.71	0.87	1.01 (0.96, 0.62–1.58) [0.79–1.37]
Mean radiation dose to tumor base (Gy), mean (median, range) [5th–95th percentile]	174 (172, 127–313) [137–205]	185 (177, 124–270) [134–264]	203 (190, 120–384) [120–363]	0.37	0.73	183 (173, 120–384) [130–266]
Radiation dose to fovea (point dose) (Gy), mean (median, range) [5th–95th percentile]	36 (23, 6–123) [11–102]	86 (80, 19–240) [19–185]	93 (81, 15–309) [33–237]	<**0.001**	0.86	63 (41, 6–309) [13–177]
Mean radiation dose to optic disc (Gy), mean (median, range) [5th–95th percentile]	24 (17, 7–78) [11–52]	54 (46, 13–139) [20–106]	84 (59, 25–400) [26–244]	<**0.001**	0.62	46 (33, 7–400) [12–110]
Max radiation dose to optic disc (Gy), mean (median, range) [5th–95th percentile]	30 (22, 8–105) [13–66]	70 (58, 15–173) [24–146]	105 (71, 30–469) [33–294]	<**0.001**	0.64	58 (42, 8–469) [14–144]
Additional treatment						
Transpupillary thermotherapy (TTT) at the time of plaque removal	10 (27)	9 (35)	4 (27)	0.80	0.73	23 (29)
Intravitreal anti-VEGF for radiation side effects	26 (70)	22 (85)	10 (67)	0.30	0.25	58 (74)

Treatment features.
Bold values indicate a significant *P* value. The initial *P* value compares all 3 groups.
*Second *P* value directly compares the latter 2 groups with optic neuropathy characterized by pallor alone or pallor with additional neuroretinal rim thinning.
Gy, gray; NA, not applicable; VEGF, vascular endothelial growth factor.

TABLE 4. Risk factors for postradiation optic atrophy after plaque radiotherapy for uveal melanoma

Outcomes	Postradiation Optic Atrophy (n = 37) [N (%)]	Pallor Only (n = 26) [N (%)]	Neuroretinal Rim Thinning and Pallor (n = 15) [N (%)]	*P* Value	*P* Value*	Total (N = 78) [N (%)]
Follow-up time (mo), mean (median, range)	73 (76, 38–121)	76 (71, 36–141)	100 (101, 46–136)	**0.01**	**0.02**	79 (78, 36–141)
Enucleation	0 (0)	3 (12)	0 (0)	**0.04**	0.29	3 (4)
Snellen visual acuity	n = 37	n = 23	n = 15			N = 75
20/20–20/40	19 (51)	0 (0)	1 (7)	<**0.001**	0.39	20 (27)
20/50–20/150	7 (19)	6 (26)	1 (7)	0.36	0.21	14 (19)
20/200 or worse	11 (30)	17 (74)	13 (87)	<**0.001**	0.44	41 (55)
Visual acuity (Snellen equivalent), mean (median, range)	20/100 (20/40, 20/20–HM)	CF (20/400, 20/50–LP)	CF (CF, 20/40–LP)	<**0.001**	0.52	20/400 (20/200, 20/20–LP)
Visual acuity (logMAR), mean (median, range)	0.65 (0.30, 0.00–3.00)	1.77 (1.30, 0.40–4.00)	2.00 (2.00, 0.30–4.00)			1.26 (1.00, 0.00–4.00)
Tumor features						
Tumor thickness (mm), mean (median, range)	1.4 (1.1, 0.0–6.1)	1.6 (0.9, 0.0–6.4)	1.3 (0.9, 0.0–4.0)	0.85	0.46	1.4 (1.0, 0.0–6.4)
Recurrence	n = 37	n = 26	n = 15			N = 78
Recurrence after plaque	0 (0)	2 (8)	0 (0)	0.14	0.52	2 (3)
Radiation complications						
Scleral necrosis	0 (0)	0 (0)	1 (7)	0.19	0.37	1 (1)
Radiation maculopathy	29 (78)	26 (100)	14 (93)	**0.01**	0.37	69 (88)
Time to radiation maculopathy (mo), mean (median, range)	32 (31, 11–75)	26 (24, 6–78)	35 (31, 5–103)	0.38	0.37	31 (27, 5–103)
Nonproliferative radiation retinopathy (NPR)	21 (57)	24 (92)	13 (87)	**0.003**	0.61	58 (74)
Time to NPR (mo), mean (median, range)	42 (36, 11–100)	27 (24, 7–78)	34 (29, 5–103)	0.20	0.71	34 (29, 5–103)
Branch retinal vein occlusion	1 (3)	4 (15)	7 (47)	<**0.001**	0.06	12 (15)
Central retinal vein occlusion	0 (0)	1 (4)	0 (0)	0.53	0.99	1 (1)
Proliferative radiation retinopathy (PR)	0 (0)	4 (15)	1 (7)	**0.02**	0.64	5 (6)

(Continued)

Outcomes	Postradiation Optic Atrophy (n = 37) [N (%)]	Pallor Only (n = 26) [N (%)]	Neuroretinal Rim Thinning and Pallor (n = 15) [N (%)]	*P* Value	*P* Value*	Total (N = 78) [N (%)]
Time to PR (mo), mean (median, range)	NA	62 (65, 40–78)	31 (31, 31-31)	NA	NA	56 (61, 31–78)
Iris neovascularization	0 (0)	1 (4)	0 (0)	0.53	0.99	1 (1)
Neovascular glaucoma	0 (0)	0 (0)	0 (0)	NA	NA	0 (0)
Neovascularization of the disc (NVD)	0 (0)	1 (4)	1 (7)	0.27	0.99	2 (3)
Neovascularization elsewhere (NVE)	0 (0)	3 (12)	1 (7)	0.08	0.99	4 (5)
Radiation papillopathy	0 (0)	26 (100)	15 (100)	<**0.001**	NA	41 (54)
Time to radiation papillopathy (mo), mean (median, range)	NA	29 (23, 5–54)	30 (22, 5–82)	NA	0.59	29 (23, 5–82)
Systemic outcomes						
Metastasis	3 (8)	2 (8)	2 (13)	0.76	0.61	7 (9)
Death	0 (0)	0 (0)	0 (0)	NA	NA	0 (0)

Outcomes.
Bold values indicate a significant *P* value. The initial *P* value compares all 3 groups.
*Second *P* value directly compares the latter 2 groups with optic neuropathy characterized by pallor alone or pallor with additional neuroretinal rim thinning.
CF, count fingers; HM, hand motion; LP, light perception; NA, not applicable.

TABLE 5. Risk factors for postradiation optic atrophy after plaque radiotherapy for uveal melanoma

Predictive Factors of Any Papillopathy	OR (95% CI)	*P* value (Critical Value)	Predictive Factors of Optic Atrophy With Neuroretinal Rim Thinning	OR (95% CI)	*P* Value (Critical Value)
Univariate analysis					
Age at diagnosis (per 1 yr increase)	1.0 (1.0–1.0)	0.75 (0.09)	Age at diagnosis (per 1 yr increase)	1.0 (1.0–1.0)	0.58 (0.08)
Male sex	0.5 (0.3–1.0)	0.05 (0.06)	Male sex	0.6 (0.2–1.5)	0.25 (0.06)
Hypertension	1.2 (0.6–2.4)	0.55 (0.09)	Hypertension	0.7 (0.2–2.3)	0.51 (0.07)
Diabetes mellitus type 2	1.8 (0.5–5.8)	0.34 (0.07)	Diabetes mellitus type 2	NA	NA
Smoking history	1.1 (0.6–2.1)	0.82 (0.10)	Smoking history	1.1 (0.4–3.2)	0.87 (0.10)
Visual acuity at presentation (per 1 logMAR increase)	5.6 (2.3–14.1)	<**0.001** (0.02)	Visual acuity at presentation (per 1 logMAR increase)	9.9 (2.5–38.9)	**0.001** (0.03)
IOP at presentation (per 1 mm Hg increase)	1.1 (1.0–1.2)	**0.03** (0.05)	IOP at presentation (per 1 mm Hg increase)	1.1 (1.0–1.3)	0.11 (0.06)
IOP maximum (per 1 mm Hg increase)	1.1 (1.0–1.2)	0.06 (0.06)	IOP maximum (per 1 mm Hg increase)	1.2 (1.1–1.4)	<**0.001** (0.01)
Tumor distance to optic disc (per 1 mm decrease)	1.3 (1.2–1.5)	<**0.001** (0.02)	Tumor distance to optic disc (per 1 mm decrease)	1.4 (1.1–1.8)	**0.006** (0.04)
Tumor distance to foveola (per 1 mm decrease)	1.2 (1.1–1.3)	<**0.001** (0.02)	Tumor distance to foveola (per 1 mm decrease)	1.2 (1.0–1.5)	**0.02** (0.05)
Subfoveal subretinal fluid at presentation	3.8 (2.0–7.1)	<**0.001** (0.02)	Subfoveal subretinal fluid at presentation	7.6 (2.5–23.2)	<**0.001** (0.01)
Prescription dose (per 1 Gy increase)	1.0 (1.0–1.0)	0.31 (0.07)	Prescription dose (per 1 Gy increase)	1.0 (0.9–1.1)	0.82 (0.09)
Prescription depth (per 1 mm increase)	1.3 (1.1–1.6)	**0.002** (0.04)	Prescription depth (per 1 mm increase)	1.4 (1.1–1.8)	**0.01** (0.04)
Radiation dose to tumor apex (per 1 Gy increase)	1.0 (1.0–1.0)	0.38 (0.08)	Radiation dose to tumor apex (per 1 Gy increase)	1.0 (1.0–1.0)	0.81 (0.09)
Radiation dose rate to tumor apex (per 1 Gy/hour increase)	0.5 (0.1–2.6)	0.38 (0.08)	Radiation dose rate to tumor apex (per 1 Gy/hour increase)	0.7 (0.0–11.1)	0.81 (0.09)

(Continued)

Predictive Factors of Any Papillopathy	OR (95% CI)	*P* value (Critical Value)	Predictive Factors of Optic Atrophy With Neuroretinal Rim Thinning	OR (95% CI)	*P* Value (Critical Value)
Radiation dose to foveola (point dose) (per 1 Gy increase)	1.01 (1.01–1.02)	<**0.001** (0.02)	Radiation dose to foveola (point dose) (per 1 Gy increase)	1.01 (1.01–1.02)	**0.001** (0.03)
Mean radiation dose to optic disc (per 1 Gy increase)	1.02 (1.02–1.03)	<**0.001** (0.02)	Mean radiation dose to optic disc (per 1 Gy increase)	1.03 (1.01–1.04)	<**0.001** (0.01)
Maximum radiation dose to optic disc (per 1 Gy increase)	1.02 (1.01–1.03)	<**0.001** (0.02)	Maximum radiation dose to optic disc (per 1 Gy increase)	1.02 (1.01–1.03)	<**0.001** (0.01)
Multivariate analysis					
Mean radiation dose to optic disc (per 1 Gy increase)	1.02 (1.01–1.04)	**0.001** (0.02)	Mean radiation dose to optic disc (per 1 Gy increase)	1.03 (1.01–1.05)	**0.003** (0.04)
IOP at presentation (per 1 mm Hg increase)	1.1 (1.0–1.2)	**0.01** (0.04)	IOP maximum (per 1 mm Hg increase)	1.4 (1.2–1.7)	<**0.001** (0.02)
Subfoveal subretinal fluid at presentation	2.0 (1.0–4.2)	**0.05** (0.06)	Subfoveal subretinal fluid at presentation	12 (2–63)	**0.004** (0.06)

Predictors of optic atrophy by Cox regression analysis.

Bold values indicate a significant *P* value. Critical values were calculated using the Benjamini–Hochberg procedure with a false discovery rate of 0.10.

For any type of optic atrophy, the model for multivariate analysis included mean radiation dose to optic disc, IOP at presentation, subfoveal subretinal fluid, prescription depth, and radiation dose to foveola.

For optic atrophy with neuroretinal rim thinning, the model for multivariate analysis included mean radiation dose to optic disc, IOP maximum, subfoveal subretinal fluid, prescription depth, and radiation dose to foveola.

CI, confidence interval; Gy, gray; IOP, intraocular pressure; OR, odds ratio.

novel modifiable risk factors for this side effect. Previous studies have identified risk factors for PROA, including radiation dose to optic disc, older patient age, diabetes mellitus, hypertension, smoking, mushroom-shaped tumor, and nasal tumor location (9,14,16,20–22). In our study, we confirmed previously identified risk factors of tumor location close to the optic disc and greater radiation dose to the disc, and we identified an additional factor of higher IOP at presentation. On subanalysis, higher maximum IOP after plaque was a risk factor for PROA characterized by both pallor and NRT.

In general, optic neuropathies are subdivided clinically into those that result in optic disc cupping/NRT or those that result in optic disc pallor. Cupping of the optic nerve head is associated with glaucomatous optic neuropathy, whereas pallor is associated with nonglaucomatous optic neuropathy. However, nonglaucomatous optic neuropathies including arteritic and nonarteritic anterior ischemic optic neuropathy (23), posterior ischemic optic neuropathy (24), compressive optic neuropathy (25), and optic neuritis (26) have rarely been associated with NRT (27). Our study adds PROA to the list of optic neuropathies that may result in NRT. In the hypothesis put forth by Claude Burgoyne et al (28), the defining events that lead to cupping and glaucomatous optic neuropathy involve deformation, remodeling, and mechanical failure of the optic nerve head connective tissue. Therefore, "glaucomatous cupping" results from IOP-related connective tissue stress and strain. The extent to which optic nerve head stress is affected by IOP is related to the vascular supply and the astrocyte and glial cell populations. On multivariate analysis in our study, the only significant predictors for PROA were greater radiation dose to optic disc, higher baseline IOP, and subfoveal subretinal fluid. Higher maximum IOP contributed to risk for concomitant NRT. This finding is in line with the hypothesis that the connective tissue deformation in any optic neuropathy may lead to NRT if the IOP at which the insult occurred is too high for the connective tissues in their present condition (28). Of note, the pressure-sensitive nature of glaucomatous optic neuropathy is not based strictly on an absolute IOP value in an individual because it relates to normative IOP values across a large population. For example, normotensive glaucoma (low-tension glaucoma) is defined as glaucomatous optic neuropathy with no history of IOP >22 mm Hg. This condition represents 30%–92% of all patients with open-angle glaucoma (29). Therefore, our observations that IOP values, although not >22, are associated with NRT is not inconsistent with the current understanding of glaucomatous optic neuropathy. Although this may not represent glaucomatous optic neuropathy, this observation suggests a pressure-sensitivity component to PROA.

In our study, compared with patients without NRT, patients with pallor and NRT had 3 mm Hg higher maximum IOP after plaque, with an associated 1.4-fold increased odds ratio of developing this subtype of PROA per each 1 mm Hg IOP increase. According to the Early Manifest Glaucoma Trial, each 1 mm Hg increment of IOP reduction was associated with a 10% reduced risk of glaucomatous optic neuropathy progression (30). These results suggest that any measurable changes in IOP could be clinically significant. Furthermore, Kiuichi et al (31) studied 1,640 Japanese atomic bomb survivors and found glaucoma in 196 patients (12%). The most common glaucoma subtype was normal-tension glaucoma, and the risk for normal-tension glaucoma correlated with higher radiation dose (31).

In our study, Drance hemorrhages, which are associated with glaucomatous optic neuropathy (32) but can be seen in other conditions such as posterior vitreous detachment or papilledema (33), were observed more frequently in the PROA group. Of note, Drance hemorrhages did not become statistically different until 36 months after plaque, whereas pallor and NRT were seen as early as 6 months after plaque. Kee et al (33) suggested the sequence of events for a disc hemorrhage results from inciting events of reactive gliosis, glial scar formation, and ultimately tractional forces from the glial scar that disrupt capillaries residing at the border of healthy and diseased retinal nerve fiber layer. In general, our observations are consistent with this idea although this hypothesis as proposed was specific to reactive gliosis in glaucoma. Because ionizing radiation (34) also induces gliosis, it is unclear whether the Drance hemorrhages observed in our study are directly related to glaucomatous optic neuropathy, gliosis from radiation, or vascular changes. Vascular changes resulting in Drance hemorrhages have also been detected by OCT angiography (OCTA) (35). Skalet et al (36), in a recent study using OCTA, found reduced peripapillary retinal capillary density (PPCD) in eyes after treatment with plaque radiotherapy compared with the untreated fellow eye, and PPCD correlated with higher radiation dose to the optic disc and worse visual acuity outcome. Chien et al (37) found that a radial peripapillary capillary density reduction could be detected by OCTA before any clinical evidence of PROA or reduction in circumpapillary retinal nerve fiber layer thickness. Reduction in peripapillary capillary density has been observed in both glaucomatous (38) and nonglaucomatous optic neuropathy (39).

Study limitations include the retrospective review of photographs and IOP measurements and the low number of patients in each group. We recognize that the group of patients with both optic nerve pallor and NRT had a longer median follow-up. However, all groups had a median follow-up longer than 5 years, with a median time to PROA less than 2 years, consistent with previous reports finding PROA most often within the first 2–3 years after treatment (8,14,17). We also acknowledge the possibility of normal-tension glaucoma producing NRT during the study period, but no fellow eyes developed glaucoma or NRT during the study period suggesting that the NRT was associated with local radiation exposure. Although not all patients had data available for each study time point, all eyes had a least 1 time point at 3, 5, or 10 years. Drance hemorrhage, disc edema, or intermittently elevated IOP could have been missed between office visits. Although photographs were compared with same day fellow eye photographs with matched phakic or pseudophakic status, judgment of pallor was admittedly subjective. We suspect the higher frequency of radiation side effects compared with previous studies (8,20) could be due to the detailed photographic review for even subtle signs of radiation side effect in our series. The exact number of anti-VEGF injections administered for each patient could not be precisely determined because many injections were given by local providers outside of our center. Finally, these patients were not examined with visual field testing or retinal nerve fiber layer or retinal ganglion cell OCT and did not have central corneal thickness or hysteresis measurements. Such additional tests might provide clarity regarding the nature of PROA and could suggest a mechanism of either glaucomatous or nonglaucomatous NRT in those patients with NRT. Future prospective studies could address these limitations.

In summary, we reviewed risk factors for PROA after plaque radiotherapy for uveal melanoma. Based on our clinical observations that some patients with PROA manifest with NRT, we here characterized PROA into 2 distinct groups, those with pallor alone and those with pallor and NRT. Based on this, we elected to evaluate glaucoma-associated metrics in the study of risk factors of PROA. On review of all types of PROA, multivariate analysis demonstrated significant risk factors of higher radiation dose to optic disc, higher baseline IOP, and subfoveal subretinal fluid. On subanalysis, eyes with both pallor and NRT had higher maximum IOP after plaque, and maximum IOP was a risk factor for this subset of PROA by multivariate Cox regression analysis. Future studies should further investigate the relationship between IOP and PROA to determine whether IOP could be a modifiable risk factor for this side effect.

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